

Curriculum Vitae

Sung-Hwan Choi, D.D.S., Ph.D.

- Professor, Department of Orthodontics, Institute of Craniofacial Deformity, Yonsei University College of Dentistry, Seoul, Republic of Korea
- Member, The Young Academy of Science and Technology of Korea (Y-KAST)

Contact Information

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Education

- **Mar. 2002 – Feb. 2008**
Doctor of Dental Surgery (D.D.S.), Dentistry
Yonsei University College of Dentistry, Seoul, Republic of Korea
- **Sep. 2009 – Aug. 2011**
Master of Science (M.S.), Orthodontics
Graduate School, Yonsei University, Seoul, Republic of Korea
- **Sep. 2014 – Feb. 2017**
Doctor of Philosophy (Ph.D.), Orthodontics
Graduate School, Yonsei University, Seoul, Republic of Korea

Positions and Employment

- **2015 – 2017**
Lecturer, Department of Orthodontics and Institute of Craniofacial Deformity, Yonsei University College of Dentistry, Seoul, Republic of Korea
- **2017 – 2018**
Clinical Assistant Professor, Department of Orthodontics and Institute of Craniofacial Deformity, Yonsei University College of Dentistry, Seoul, Republic of Korea
- **2018 – 2022**
Assistant Professor, Department of Orthodontics and Institute of Craniofacial Deformity, Yonsei University College of Dentistry, Seoul, Republic of Korea

- **2022 – 2025**
Associate Professor, Department of Orthodontics and Institute of Craniofacial Deformity, Yonsei University College of Dentistry, Seoul, Republic of Korea
- **2025 – Present**
Professor, Department of Orthodontics and Institute of Craniofacial Deformity, Yonsei University College of Dentistry, Seoul, Republic of Korea
- **2025 – 2026**
Visiting Professor, Center for Systems Biology, Massachusetts General Hospital / Harvard Medical School, MA, USA

Administrative Appointments

- **2022 – 2025**
Associate Dean of Student Affairs, Yonsei University College of Dentistry, Seoul, Republic of Korea

Honors and Awards

- **2017**
Young Researcher Award, Korean Dental Association, Seoul, Republic of Korea
- **2018**
Young Researcher Award, Korean Association of Orthodontists, Seoul, Republic of Korea
- **2019**
Minister's Commendation for Young Researcher Achievement, Ministry of Health and Welfare, Republic of Korea
- **2021**
The 17th Yeonsong Dental Award (Gold Prize), Korean Academy of Dental Science, Seoul, Republic of Korea
- **2023**
The 3rd MINEC Academic Award (Grand Prize), Korean Academy of Dental Science, Seoul, Republic of Korea
- **2024**
Elected Member, The Young Academy of Science and Technology of Korea (Y-KAST)

Research Grants and Support

As Principal Investigator (PI)

- **2018 – 2021**

Project Title: Research on Protein- and Biofilm-Repellent Properties in Dental Biomaterials Using Zwitterions

Funding Agency: National Research Foundation of Korea (NRF), Republic of Korea

Role: Principal Investigator (PI)

- **2020 – 2022**

Project Title: Development of Core Technology for Dental iPMMA to Suppress Biofilm Adhesion

Funding Agency: Korea Medical Device Development Fund (KMDF), Republic of Korea

Role: Principal Investigator (PI)

- **2021 – 2026**

Project Title: Research on Antifouling and Bioactive Dental Biomaterials Using a Zwitterionic Network Modifier

Funding Agency: National Research Foundation of Korea (NRF), Republic of Korea

Role: Principal Investigator (PI)

- **2023 – 2025**

Project Title: Development of Biofilm-Repellent Dental CAD/CAM Blocks and Anti-Cariogenic Bioactive Cements

Funding Agency: Korea Medical Device Development Fund (KMDF), Republic of Korea

Role: Principal Investigator (PI)

- **2023 – Present**

Project Title: Development of Symbiotic Biocompatible Medical Device Materials and Productization Technology for Patient-Customized Direct-Printed Biliary Stents and Orthodontic Appliances

Funding Agency: Ministry of Trade, Industry and Energy (MOTIE), Republic of Korea

Role: Principal Investigator (PI)

As Co-Principal Investigator (Co-PI)

- **2023 – 2026**

Project Title: Research Center for the Autonomic Nervous System and Bone Homeostasis Project

Funding Agency: National Research Foundation of Korea (NRF), Republic of Korea

Role: Co-Principal Investigator (Co-PI)

- **2025 – Present**

Project Title: Development of Commercialized Convergence Medical Products for Oral Disease Treatment

Funding Agency: Regional Innovation System & Education (RISE) Program through the Gangwon

RISE Center, funded by the Ministry of Education (MOE) and Gangwon State, Republic of Korea

Role: Co-Principal Investigator (Co-PI)

Peer-Reviewed Publications

Selected High-Impact Publications (As Corresponding Author)

1. Choi W, Mangal U, Cha JK, Cho H, Ryu JH, Kim JY, Koh WG, Lee KJ, Kim KW, **Choi SH***, Traverso G, Hong J. "A Mucosa-Inspired Dynamic Biointerface Engineering a Biofilm-Preventive Niche against Pathogenic Microbiome Dysbiosis." *Advanced Materials*. 2026. In Press (Accepted). **(IF: 29.1, JCR Top 2%)**
2. Lee MY, Mangal U, Yoon HW, Im H, **Choi SH***, Kwon JS, Shin SJ. "Microfluidic pulp platform with vascular integration to evaluate biofunctional material." *Biomaterials*. 2026;334:124221. **(IF: 13.6, JCR Top 6%)**
3. Choi W, Mangal U, Yu JH, Ryu JH, Kim JY, Jun T, Lee Y, Cho H, Choi M, Lee M, Ryu DY, Lee SY, Jung SY, Cha JK, Cha JY, Lee KJ, Lee S, **Choi SH***, Hong J. "Viscoelastic and antimicrobial dental care bioplastic with recyclable life cycle." *Nature Communications*. 2024;15(1):9205. **(IF: 18.1, JCR Top 6%)**
4. Ryu JH, Mangal U, Kwon JS, Seo JY, Byun SY, Lee YH, Jang S, Hwang G, Ku H, Shin Y, Kim D, **Choi SH***. "Integrating Phosphate Enhances Biomineralization Effect of Methacrylate Cement in Vital Pulp Treatment with Improved Human Dental Pulp Stem Cells Stimulation." *Advanced Healthcare Materials*. 2024;13(31):e2402397. **(IF: 11.0, JCR Top 9%)**
5. Wu C, Mangal U, Seo JY, Kim H, Bai N, Cha JY, Lee KJ, Kwon JS, **Choi SH***. "Enhancing biofilm resistance and preserving optical translucency of 3D printed clear aligners through carboxybetaine-copolymer surface treatment." *Dental Materials*. 2024;40(10):1575-1583. **(IF: 6.3, JCR Top 4%)**
6. Jung S, Yong H, Lee M, Kim B, Kang TY, Jang HS, Choi WJ, Kim D, Choi M, Bang G, Hwang H, **Choi SH***, Hong J, Lee S. "Unraveling the Missing Link of Bio-Electrical Stimulation from Body-Mediated Energy Transfer." *Advanced Functional Materials*. 2023;33(36):2302465. **(IF: 19.9, JCR Top 5%)**
7. Choi W, Mangal U, Park JY, Kim JY, Jun T, Jung JW, Choi M, Jung S, Lee M, Na JY, Ryu DY, Kim JM, Kwon JS, Koh WG, Lee S, Hwang PTJ, Lee KJ, Jung UW, Cha JK, **Choi SH***, Hong J. "Occlusive membranes for guided regeneration of inflamed tissue defects." *Nature Communications*. 2023;14(1):7687. **(IF: 18.1, JCR Top 6%)**
8. Jin J, Mangal U, Seo JY, Kim JY, Ryu JH, Lee YH, Lugtu C, Hwang G, Cha JY, Lee KJ, Yu HS, Kim KM, Jang S, Kwon JS, **Choi SH***. "Cerium oxide nanozymes confer a cytoprotective and bio-friendly surface micro-environment to methacrylate based oro-facial prostheses." *Biomaterials*.

2023;296:122063. (IF: 13.6, JCR Top 6%)

9. Kim JY, Choi W, Mangal U, Seo JY, Kang TY, Lee J, Kim T, Cha JY, Lee KJ, Kim KM, Kim JM, Kim D, Kwon JS, Hong J, **Choi SH***. "Multivalent network modifier upregulates bioactivity of multispecies biofilm-resistant polyalkenoate cement." *Bioactive Materials*. 2022;14:219-233. (IF: 23.6, JCR Top 2%)
10. Choi W, Park S, Kwon JS, Jang EY, Kim JY, Heo J, Hwang Y, Kim BS, Moon JH, Jung S, **Choi SH***, Lee H, Ahn HW, Hong J. "Reverse Actuation of Polyelectrolyte Effect for In Vivo Antifouling." *ACS Nano*. 2021;15(4):6811-6828. (IF: 17.3, JCR Top 6%)
11. Lee MJ, Kwon JS, Kim JY, Ryu JH, Seo JY, Jang S, Kim KM, Hwang CJ, **Choi SH***. Bioactive resin-based composite with surface pre-reacted glass-ionomer filler and zwitterionic material to prevent the formation of multi-species biofilm. *Dental Materials*. 2019;35(9):1331-1341. (IF: 6.3, JCR Top 4%)

Selected Peer-Reviewed Publications (As First or Corresponding Author)

2026

1. Choi EA, Shim EA, Youn JH, Mangal U, Kim H, **Choi SH***. Skeletal and biological predictors of chin soft-tissue thickness and curvature radius in Korean pediatric patients: A cone-beam computed tomography and 3-dimensional facial scanning study. *Am J Orthod Dentofacial Orthop*. 2026. In Press (Accepted).
2. Kim MJ, Seo JY, Mangal U, **Choi SH***. Inhibition of cariogenic biofilms by a zinc-doped phosphate-based glass orthodontic adhesive. *BMC Oral Health*. 2026. In Press (Accepted).
3. Ryu JH, Mangal U, Kim JH, Kwon JS, Kim KW, **Choi SH***. Dose-dependent effects of zinc oxide nanoparticles on pre-osteoblast cellular response. *Sci Rep*. 2026. In Press (Accepted).
4. Yu JH, Mangal U, Liu J, Kim JH, Yu HS, Lee KJ, Kwon JS, **Choi SH***, Cha JY. Influence of restorative material types on 3D printed aligner accuracy using intraoral scanners. *BMC Oral Health*. 2026. In Press (Accepted).
5. Mangal U, Jinn H, Hyun CM, Kim H, Cha JY, Lee KJ, **Choi SH***. Facial surface tracing approach to reduce the observation errors in 3D curvilinear landmark annotations on cone-beam computed tomography images. *J Dent*. 2026;168:106575.
6. Kim GT, Mangal U, Yu JH, Kim JE, Hwang G, Kwon JS, **Choi SH***. Comparative Analysis of the Mechanical, Surface, and Dimensional Properties of 3D-Printed Aligners: Influence of Printer Type and Nitrogen-Assisted Post-Curing. *Orthod Craniofac Res*. 2026. In Press (Accepted).
7. Ryu JH, Byun SY, Mangal U, Kim JH, Yoo J, Kim KW, Kwon JS, **Choi SH***. Calcium silicate cement with phosphate-based glass for enhanced osteogenic activity and mechanical strength. *Clin*

Oral Investig. 2026;30(5):187.

8. Choi EA, Shin Y, Yu JH, Kim J, Mangal U, Kim JY, **Choi SH***. Clinical acceptability and repeatability of three-dimensional maxillary positioning following orthognathic surgical wafer placement: An in vitro study. *J Dent.* 2026;167:106536.
9. Andrew Choi EH, Mangal U, Ryu JH, Kim JH, Hwang G, Kim H, Cha JY, Lee KJ, **Choi SH***. Changes in the cytocompatibility, surface hardness, and surface topography of 3-dimensional-printed clear aligners after clinical use. *Am J Orthod Dentofacial Orthop.* 2026;169(2):168-176.
10. Park YS, Andrew Choi EH, Jamal H Alwaznah F, Kim H, Choi YJ, Lee KJ, Yu HS, **Choi SH***. Is apex position a determining factor in the orthodontic traction duration of unilateral impacted maxillary canines? A cone-beam computed tomography-based 3-dimensional study. *Am J Orthod Dentofacial Orthop.* 2026;169(2):168-176.

2025

11. Wu C, Mangal U, Bai N, Kim H, Hwang G, Cha JY, Lee KJ, Kwon JS, **Choi SH***. Effects of printing layer thickness and build orientation on the mechanical properties and color stability of 3D-printed clear aligners. *Angle Orthod.* 2025;95(4):355-361.
12. Choi W, Mangal U, Park H, Lee M, Tran NQ, Lee KJ, Thi PL, Kimm H, **Choi SH***, Hong J. Biodegradation of β -Sheet Crystal-Rich Silk Fibroin Bioplastic Promoted by Functionally Active Soil Microbiome. *ACS EST Engg.* 2025;5(11):3423-3433.
13. Ryu JH, Mangal U, Kim JH, Yang DJ, Lee KJ, Choi YH, Shin DM, **Choi SH***, Kim KW. SHLP2 restores pre-osteoblastic cells against oxidative stress-induced inflammaging. *Sci Rep.* 2025;16(1):938.
14. Kim YN, Choi EA, Mangal U, Yu JH, Cha JY, Park SH, Kim JY, **Choi SH***, Yu HS. Soft tissue thickness changes in the lower face following bimaxillary surgery with advancement genioplasty in skeletal Class III malocclusion. *BMC Oral Health.* 2025;26(1):45.
15. Kim MJ, Kim E, Mangal U, Seo JY, Cha JY, Lee KJ, Choi YJ, Kwon JS, Yu HS, **Choi SH***. Biofilm-resistant zwitterionic resin-based adhesive for orthodontic bracket-tooth interfaces: an in vitro evaluation. *Clin Oral Investig.* 2025;29(10):478.

2024

16. Choi EA, Bae JH, Lee S, Kim JH, Nguyen LP, Kim JY, **Choi SH***. Multidisciplinary approach to occlusal rehabilitation in a patient with true hemifacial hyperplasia and temporomandibular joint ankylosis: a case report. *BMC Oral Health.* 2024;24(1):1525.
17. Bae JH, Mangal U, Yu JH, Cha JY, Yu HS, Lee KJ, **Choi SH***. Retention efficacy and patient experience with customized clear retainer and wrap-around circumferential retainer among non-

extraction orthodontic patients: A 12-month follow-up cohort study. *Orthod Craniofac Res.* 2024;27(6):928-937.

18. Nam SH, Cha JY, Lee KJ, Lee SH, **Choi SH***, Yu HS. Factors affecting unusual postoperative vertical downward movement of mandible after surgery-first approach using intraoral vertical ramus osteotomy. *BMC Oral Health.* 2024;24(1):1203.
19. Kim MJ, Mangal U, Seo JY, Kim JY, Kim JY, Ryu JH, Kim HJ, Lee KJ, Kwon JS, **Choi SH***. A novel zwitterion incorporated Nano-crystalline ceramic and polymer for bacterial resistant dental CAD-CAM block. *J Dent.* 2024;148:105054.
20. Kim JE, Mangal U, Yu JH, Kim GT, Kim H, Seo JY, Cha JY, Lee KJ, Kwon JS, **Choi SH***. Evaluation of the effects of temperature and centrifugation time on elimination of uncured resin from 3D-printed dental aligners. *Sci Rep.* 2024;14(1):15206.
21. Ryu JH, Mangal U, Yoo J, Youm JH, Kim JY, Seo JY, Kim D, Kwon JS, **Choi SH***. Low concentration zinc oxide nanoparticles enrichment enhances bacterial and pro-inflammatory resistance of calcium silicate-based cements. *J Mech Behav Biomed Mater.* 2024;151:106399.

2023

22. Wu C, Mangal U, Kim J, Lee KJ, Cha JY, Kwon JS, **Choi SH***. Quantitative light-induced fluorescence enables effective detection of orthodontic adhesive residues in diverse environments. *Photodiagnosis Photodyn Ther.* 2023;44:103743.
23. Kim MJ, Seo JY, Jung IJ, Mangal U, Kim HJ, Lee KJ, Lee MJ, Kwon JS, **Choi SH***. A novel orthodontic adhesive containing zinc-doped phosphate-based glass for preventing white spot lesions. *J Dent.* 2023;137:104689.
24. Ryu JH, Mangal U, Lee MJ, Seo JY, Jeong IJ, Park JY, Na JY, Lee KJ, Yu HS, Cha JK, Kwon JS, **Choi SH***. Effect of strontium substitution on functional activity of phosphate-based glass. *Biomater Sci.* 2023;11(18):6299-6310.
25. Kim H, Kim CS, Lee JM, Lee JJ, Lee J, Kim JS, **Choi SH***. Prediction of Fishman's skeletal maturity indicators using artificial intelligence. *Sci Rep.* 2023;13(1):5870.
26. Mangal U, Noh K, Lee S, Cha JK, Song JS, Cha JY, Lee KJ, Kim KM, Kwon JS, **Choi SH***. Multistability and hysteresis in states of oral microbiota: Is it impacting the dental clinical cohort studies? *J Periodontal Res.* 2023;58(2):381-391.
27. Wu C, Kim MJ, Mangal U, Seo JY, Kim JY, Kim J, Park JY, Kwon JS, **Choi SH***. Effect of bacterial resistant zwitterionic derivative incorporation on the physical properties of resin-modified glass ionomer luting cement. *Sci Rep.* 2023;13(1):3589.
28. Mangal U, Lee SM, Lee S, Cha JY, Lee KJ, Yu HS, Jung H, **Choi SH***. Reorientation methodology

for reproducible head posture in serial cone beam computed tomography images. *Sci Rep.* 2023;13(1):3220.

29. Lee Y, Lee JM, Park SH, Choi YJ, **Choi SH***, Hwang JJ, Yu HS. Three-dimensional soft tissue landmark detection with marching cube algorithm. *Sci Rep.* 2023;13(1):1544.
30. Kim H, Kim JS, Kim CS, Becker-Weimann SY, Cha JY, **Choi SH***. Skin irritation in children undergoing orthodontic facemask therapy. *Sci Rep.* 2023;13(1):2200.
31. Mangal U, Seo JY, Ryu JH, Jin J, Wu C, Cha JY, Lee KJ, Yu HS, Kim KM, Kwon JS, **Choi SH***. Changes in mechanical and bacterial properties of denture base resin following nanoceria incorporation with and without SBA-15 carriers. *J Mech Behav Biomed Mater.* 2023;138:105634.
32. Mangal U, Kang TY, Jung JW, Kim JY, Seo JY, Cha JY, Lee KJ, Yu HS, Kim KM, Kim JM, Kwon JS, **Choi SH***. Polybetaine-enhanced hybrid ionomer cement shows improved total biological effect with bacterial resistance and cellular stimulation. *Biomater Sci.* 2023;11(2):554-566.

2020 – 2022

33. Lee MJ, Kim MJ, Mangal U, Seo JY, Kwon JS, **Choi SH***. Zinc-modified phosphate-based glass micro-filler improves *Candida albicans* resistance of auto-polymerized acrylic resin without altering mechanical performance. *Sci Rep.* 2022;12(1):19456.
34. Yang JS, Cha JY, Lee JY, **Choi SH***. Radiographical characteristics and traction duration of impacted maxillary canine requiring surgical exposure and orthodontic traction: a cross-sectional study. *Sci Rep.* 2022;12(1):19183.
35. Jin J, Bhat R, Mangal U, Seo JY, Min Y, Yu J, Kim DE, Kuroda K, Kwon JS, **Choi SH***. Molecular weight tuning optimizes poly(2-methoxyethyl acrylate) dispersion to enhance the aging resistance and anti-fouling behavior of denture base resin. *Biomater Sci.* 2022;10(9):2224-2236.
36. Mangal U, Arum H, Huisoo K, Jung YH, Lee KJ, Yu HS, Hwang JJ, **Choi SH***. Tomographic similarity scan with a computed modified absolute mandibular midsagittal plane for precise and objective localization of mandibular asymmetry. *Comput Biol Med.* 2021;134:104465.
37. **Choi SH**, Jeon JY, Lee KJ, Hwang CJ. Clinical applications of miniscrews that broaden the scope of non-surgical orthodontic treatment. *Orthod Craniofac Res.* 2021;24 Suppl 1:48-58.
38. **Choi SH**, Shin J, Cha JK, Kwon JS, Cha JY, Hwang CJ. Evaluation of success rate and biomechanical stability of ultraviolet-photofunctionalized miniscrews with short lengths. *Am J Orthod Dentofacial Orthop.* 2021;159(2):158-166.
39. **Choi SH**, Lee H, Hwang JJ, Jung HD, Hwang CJ, Cha JY. Differences in soft-tissue thickness changes after bimaxillary surgery between patients with vertically high angle and normal angle. *Am J Orthod Dentofacial Orthop.* 2021;159(1):30-40.

40. Mangal U, Min YJ, Seo JY, Kim DE, Cha JY, Lee KJ, Kwon JS, **Choi SH***. Changes in tribological and antibacterial properties of poly(methyl methacrylate)-based 3D-printed intra-oral appliances by incorporating nanodiamonds. *J Mech Behav Biomed Mater.* 2020;110:103992.
41. Baik WK, Baik HS, **Choi SH***. Comprehensive orthodontic treatment of a young girl with an odontogenic keratocyst and impacted teeth in the mandible. *Korean J Orthod.* 2020;50(1):63-71.

2015 – 2019

42. Kim D, Lee MJ, Kim JY, Lee D, Kwon JS, **Choi SH***. Incorporation of zwitterionic materials into light-curable fluoride varnish for biofilm inhibition and caries prevention. *Sci Rep.* 2019;9(1):19550.
43. **Choi SH**, Ryu JH, Kwon JS, Kim JE, Cha JY, Lee KJ, Yu HS, Choi EH, Kim KM, Hwang CJ. Effect of wet storage on the bioactivity of ultraviolet light- and non-thermal atmospheric pressure plasma-treated titanium and zirconia implant surfaces. *Mater Sci Eng C Mater Biol Appl.* 2019;105:110049.
44. Kwon SM, Hwang JJ, Jung YH, Cho BH, Lee KJ, Hwang CJ, **Choi SH***. Similarity index for intuitive assessment of three-dimensional facial asymmetry. *Sci Rep.* 2019;9(1):10959.
45. Kwon JS, Lee MJ, Kim JY, Kim D, Ryu JH, Jang S, Kim KM, Hwang CJ, **Choi SH***. Novel anti-biofouling light-curable fluoride varnish containing 2-methacryloyloxyethyl phosphorylcholine to prevent enamel demineralization. *Sci Rep.* 2019;9(1):1432.
46. **Choi SH**, Jeong WS, Cha JY, Lee JH, Lee KJ, Yu HS, Choi EH, Kim KM, Hwang CJ. Effect of the ultraviolet light treatment and storage methods on the biological activity of a titanium implant surface. *Dent Mater.* 2017;33(12):1426-1435.
47. **Choi SH**, Jeong WS, Cha JY, Lee JH, Lee KJ, Yu HS, Choi EH, Kim KM, Hwang CJ. Overcoming the biological aging of titanium using a wet storage method after ultraviolet treatment. *Sci Rep.* 2017;7(1):3833.
48. **Choi SH**, Fan D, Hwang MS, Lee HK, Hwang CJ. Effect of growth hormone treatment on craniofacial growth in children: Idiopathic short stature versus growth hormone deficiency. *J Formos Med Assoc.* 2017;116(4):313-321.
49. **Choi SH**, Kim JS, Kim CS, Yu HS, Hwang CJ. Cone-beam computed tomography for the assessment of root-crown ratios of the maxillary and mandibular incisors in a Korean population. *Korean J Orthod.* 2017;47(1):39-49.
50. **Choi SH**, Jung SN, Cha JY, Hu KS, Kim KD, Baik HS, Hwang CJ. Changes in the craniofacial complex and alveolar bone height of young adults: Applications to dental medicine. *Clin Anat.* 2016;29(8):1011-1017.
51. **Choi SH**, Kang DY, Cha JY, Jung YS, Baik HS, Hwang CJ. Is There a Difference in Stability After Intraoral Vertical Ramus Osteotomy Between Vertically High-Angle and Normal-Angle Patients? *J*

Oral Maxillofac Surg. 2016;74(11):2252-2260.

52. **Choi SH**, Yoo HJ, Lee JY, Jung YS, Choi JW, Lee KJ. Stability of pre-orthodontic orthognathic surgery depending on mandibular surgical techniques: SSRO vs IVRO. ***J Craniomaxillofac Surg.*** 2016;44(9):1209-15.
53. **Choi SH**, Shi KK, Cha JY, Park YC, Lee KJ. Nonsurgical miniscrew-assisted rapid maxillary expansion results in acceptable stability in young adults. ***Angle Orthod.*** 2016;86(5):713-20.
54. **Choi SH**, Kim SJ, Lee KJ, Sung SJ, Chun YS, Hwang CJ. Stress distributions in peri-miniscrew areas from cylindrical and tapered miniscrews inserted at different angles. ***Korean J Orthod.*** 2016;46(4):189-98.
55. **Choi SH**, Kim YH, Lee KJ, Hwang CJ. Effect of labiolingual inclination of a maxillary central incisor and surrounding alveolar bone loss on periodontal stress: A finite element analysis. ***Korean J Orthod.*** 2016;46(3):155-62.
56. **Choi SH**, Kang DY, Cha JY, Jung YS, Yu HS, Park HS, Hwang CJ. Major factors contributing to anterior and posterior relapse after intraoral vertical ramus osteotomy. ***J Craniomaxillofac Surg.*** 2016;44(4):413-20.
57. **Choi SH**, Cha JY, Park HS, Hwang CJ. Intraoral Vertical Ramus Osteotomy Results in Good Long-Term Mandibular Stability in Patients With Mandibular Prognathism and Anterior Open Bite. ***J Oral Maxillofac Surg.*** 2016;74(4):804-10.
58. **Choi SH**, Kim JS, Cha JY, Hwang CJ. Effect of malocclusion severity on oral health-related quality of life and food intake ability in a Korean population. ***Am J Orthod Dentofacial Orthop.*** 2016;149(3):384-90.
59. **Choi SH**, Hwang CJ, Baik HS, Jung YS, Lee KJ. Stability of Pre-Orthodontic Orthognathic Surgery Using Intraoral Vertical Ramus Osteotomy Versus Conventional Treatment. ***J Oral Maxillofac Surg.*** 2016;74(3):610-9.
60. **Choi SH**, Jang SH, Cha JY, Hwang CJ. Evaluation of the surface characteristics of anodic oxidized miniscrews and their impact on biomechanical stability: An experimental study in beagle dogs. ***Am J Orthod Dentofacial Orthop.*** 2016;149(1):31-8.
61. **Choi SH**, Won JH, Cha JY, Hwang CJ. Effect of Audiovisual Treatment Information on Relieving Anxiety in Patients Undergoing Impacted Mandibular Third Molar Removal. ***J Oral Maxillofac Surg.*** 2015;73(11):2087-92.
62. **Choi SH**, Kim BI, Cha JY, Hwang CJ. Impact of malocclusion and common oral diseases on oral health-related quality of life in young adults. ***Am J Orthod Dentofacial Orthop.*** 2015;147(5):587-95.